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Jefferson Williams

Senior Machine Learning Engineer

SUMMARY

Skilled machine learning professional with 7+ years of experience in designing and developing machine learning models and systems across diverse industries. Possesses a diverse experience in collating & analyzing data, enhancing existing models, and revamping business processes as part of providing compelling value to key clients. Proficient in collaborating with teams of high-performing professionals for developing machine learning frameworks and solutions for slashing costs & optimizing time for catapulting business growth.

KEY SKILLS

- Team Management & Organizational Leadership
- Communications
- Corporate Communications Management
- Consumer Behavior Analysis
- Passionate about machine learning and data
- Scientific mindset and strong motivation to learn
- Inspired by using machine learning to improve decision-making
- Quality Control

TECHNICAL SKILLS

Tensorflow | PyTorch | Keras | RNN | CNN | TNN | Scikit-learn | Pandas | Numpy | Data Analytics with R | Python | C/C++ | SQL | AWS | Git | Trello | VSCode | Jupyter

APPLICATIONS

Natural Language Process | Speech to text | Text to speech | Image captioning | Object detection | Image classification | Classification | Regression | Clustering

PROFESSIONAL EXPERIENCE

Senior Machine Learning Engineer

Dec '21 - Present

self project related to IE 4.0

Remote Work

I have to make the agent using Advantage actor-critic algorithm (Reinforcement Algorithm)

Key Achievement

- Design the architecture of agent that detect the target and finish the mission.
- Implemented the A2C algorithm for agent's brain.
- For initial stage of the agent, I have simulated this A2C algorithm in DeepMind lab environment

Summary of this project:

A2C algorithm is an advantage actor-critic algorithm for reinforcement learning.
Reinforcement Learning algorithm is trained by critic. Other machine learning algorithms such as supervised learning, unsupervised learning are required to need the dataset for training and testing.
But Reinforcement Learning is trained by critic. Critic is to provide reward for an actor's action.
By this reward, the actor is trained. But this algorithm is needed more and more time than other machine learning algorithms.
But this algorithm don't need the dataset. So using this algorithm, an agent that has the brain like a human can be implemented.
At present, Our agent is in training stage. But A2C algorithm has the same actor and value network. So Accuracy is so low. To increase the accuracy, we have to implement the pre-building A2C algorithm like Stable baseline3.
Using this algorithm, we have to check this A2C has the high accuracy in DeepMind lab environment and after that we have to research internal architecture of this stable baseline3. After that stage, we must this algorithm to our agent.

So, agent have to recognize the target and finish the mission. Without training, initially, agent don't finish the mission. But during the training, agent have to finish the mission. So I have to train the agent using A2C algorithm.

Senior Machine Learning Engineer
Mar '20 - Nov '21

deemsoft.com
Remote Work

Speech recognition and transcription service enabled with Artifical Intelligence

Key Achievement

- Led a team of 3 engineers to deploy ML/DL methodologies & effectively execute projects

Machine Learning models & Application Development

- Initiated improved products in the domain of customer supported services, real-time service, digital marketing services
- Utilized Deep Learning packages like Tensorflow, PyTorch to develop Machine Learning models
- Created models from speech data for English & developed an autonomous audio solution system
- Developing solutions to improve training and evaluation of ML models under sampling bias
- Preprocessed data and developed project specific ML pipelines and prototypes for corporate clients

Predictive Modelling & Scalable Solutions

- Deployed Predictive Modelling techniques including Machine Learning
- Processed terabytes of data via Machine Learning & Predictive Modelling to develop dynamic solutions
- Developed State of the art large-scale Machine Learning Services & Applications on the Cloud by applying predictive technology

Junior Machine Learning Engineer
Mar '19 - Dec '19

kea.ai
Remote Work

Natural Language Processing AI-Powered voice assistance

Designing and optimizing Machine Learning algorithms

- Assisted the implementation and deployment of systems used for natural language input and processing, data collection , Machine Learning model training, and deployment of machine Learning models
- Built new models to extract more value from the customer data we collect. Which will help us know more about users and merchants than anyone else in marketing or retail
- Developed high precision classifiers and tools leveraging Machine Learning, regression and rule-based models
- Used Machine Learning and statistical modeling techniques to develop and evaluate algorithms to improve performance, quality, data management and accuracy
- Worked with product manager to formulate the data analytics problem
- Designed data pipeline architecture by adhering to client's requirements and increased users by 100k within 6 months
- Communicated with project managers and brainstormed recommendations to increase efficiency by 43%

Junior Machine Learning Engineer
Mar '18 - Dec '18

Anyline.com
Remote Work

Accurate, secure mobile data capture

Designing and optimizing Machine Learning algorithms

- Developed cross-platform and efficient algorithms, service tools, models and related solutions for OCR
- Collaborated with design and technical teams in implementing Machine Learning, computer vision and image processing algorithms into well-defined architectures
- Implemented models in different software and hardware
- Working with QA personnel in executing automation tests for image processing algorithms and supporting software or applications before deployment as required as well as utilizing various design techniques, tools and principles to write technical reports and prepare user manuals for the products developed
- Developed and enhanced image processing techniques or methods for advanced systems as per user and business requirements in a real-time operating environment as well as optimized related algorithms for mobile platforms by using dedicated hardware
- Utilized various tools to support image quality assessments throughout all phases of system development as required as well as led and participated enterprise efforts to resolve image processing and quality anomalies and discrepancies

Middle Machine Learning Engineer
Mar '17 - Dec '17

nanit.com

Smart Baby Monitor-track sleep, Breathing Motion, and growth

Designing and optimizing Machine Learning algorithms

- Designed and implemented and optimized algorithms for detecting baby motion and sleeping status using deep learning(convolutional neural nets)
- Implemented a CNN+SVM anomaly detector
- Implemented convolutional neural nets to determine baby sleeping status
- Actively performed data analysis for system performance enhancement

Deep Learning Engineer
Apr '16 - Nov '16

azcapthca.com

Automated online captcha solver API service

Designing and implementing models

- Hands-on experience in software engineering and machine learning
- Designed, optimized and implemented Deep Learning models for image processing
- Developed, optimized models for image processing
- Communicated effectively with upper management through weekly meetings and project progress reports

EDUCATION

Bachelor's degree in computer science

Apr '12 - Aug '15

Florida State University

Tallahassee, FL

Focused on machine learning and data science

When I was a student, I tried to get skills such as computer science, math and statistics(knowledge of what makes machine learning algorithms work) to become a good machine learning engineer. After graduated from university, I had part time job on freelancer.com.